Matthew Ho

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Highlights

- **36 papers** in leading astrophysics, machine learning, and physics journals (18 as first or second author).
- **19 students mentored** (11 undergraduate, 5 Master's, 3 Ph.D.).
- **$7M+ in research funding** secured as a core collaborator on major grants.
- **Lead the Implicit Likelihood Inference working group** of the Simons Collaboration on Learning the Universe, building its core ML inference framework and leading its flagship cosmological analyses.
- **10+ university courses and summer schools** taught as instructor, lecturer, and teaching assistant.

Research Experience

2026 – 2029 **Member**, Institute for Advanced Study — Princeton, NJ
Research on machine learning methods for constraining astrophysical theory with observations, including simulation-based inference, accelerated forward modeling, interpretable representation learning, and agentic science.

2022 – 2026 **Postdoctoral Research Fellow**, Institut d'Astrophysique de Paris & Columbia University — Paris & NYC
Joint 2+2 year position within the Simons Collaboration on Learning the Universe developing accelerated emulators and inference models for cosmological analysis.
Advisors: Benjamin Wandelt, Guilhem Lavaux, Greg Bryan

Education

2017 – 2022 **Ph.D., M.S. Physics**, Carnegie Mellon University — Pittsburgh, PA
Thesis: *Deep Learning for Dynamical Mass Estimation of Galaxy Clusters*
Advisor: Hy Trac · GPA: 3.97

2021 – 2022 **M.S. Machine Learning**, Carnegie Mellon University — Pittsburgh, PA
Research Interests: Deep Learning, Bayesian Modeling, Approximate Inference, Generative Models · GPA: 4.0

2014 – 2017 **B.S. Engineering Physics**, University of Illinois at Urbana-Champaign — Urbana, IL
Minor in Mathematics · Research advisors: Lucas Wagner, Guy Garnett

Grants & Fellowships

2026 **Keystone Astronomy & AI (KAAI) Fellowship**, McWilliams Center, CMU
One-month visiting fellowship, supported by the Simons Foundation, conducting research on foundation models and agentic methods for astrophysical simulation, and hosting an associated workshop.

2025 – 2028 **Simons Collaboration**, Simons Foundation — **$6,000,000**
Collaborator on the renewal of the Learning the Universe Simons Collaboration ($700K to Columbia University), advancing machine learning analysis of cosmological and galaxy formation models for next-generation observatories including JWST, DESI, *Euclid*, and the Simons Observatory.

2024 – 2025 **NSF ACCESS Maximize Computing Grant**, National Science Foundation — **$438,732**
Collaborator on a computing grant of 42 million CPU hours and 20 GPU-years for dark matter and hydrodynamical simulations and machine learning emulators for the Learning the Universe project.

2023 – 2025 **ANR Appel à projets générique**, French National Research Agency (ANR) **€304,261**
Collaborator on ANR AAPG grant INFOCW, applying state-of-the-art simulations and machine learning inference to large-scale structure maps to infer cosmological parameters and initial conditions.

2022 – 2025 **NSF Astronomy and Astrophysics Research Grant**, National Science Foundation **$538,957**
Collaborator on NSF grant AST 2206055, developing multiwavelength machine learning models to probe halo environments and galaxy formation in cluster- and group-scale systems.

2021 **John Peoples, Jr. Research Fellowship in Physics**, Department of Physics, CMU
Full tuition and stipend, awarded annually to one outstanding physics graduate student.

2020 **McWilliams-PSC Research Seed Grant**, McWilliams Center for Cosmology
Full tuition and stipend for innovative, high-impact scientific research.

Publications

★ highlighted work † student mentee Up-to-date list: [ADS library]

UNDER REVIEW (8)

★ [36] **Celia: Catalog Embeddings Learned From Cosmological Simulations**
Zhang, X., **Ho, M.**, Kumar, A.†, et al. · Submitted to *OJAp* · 2026 · [ADS]

[35] **The Deep Learning Halo Definer: A Multimodal Framework for Halo Mass and Gas Fraction Inference on Galaxy Groups and Clusters**
Ogle, C.†, Poncelet Adams, K. N.†, …, **Ho, M.**, et al. · Submitted to *OJAp* · 2026 · [ADS]

[34] **CNN-Based Inference of Gaseous Halo Properties from Synthetic X-ray and 21-cm HI Observations**
Poncelet Adams, K. N.†, Oppenheimer, B. D., …, Ogle, C.†, **Ho, M.**, et al. · Submitted to *OJAp* · 2026 · [ADS]

[33] **Don't Cut Corners: How Training Outside the Prior Makes Simulation-Based Inference More Robust**
Tirapongprasert, C.† & **Ho, M.** · Submitted to *Mach. Learn.: Sci. Technol.* · 2026 · [ADS]

[32] **Cluster Mass Inference from Galaxy Kinematics**
Wang, B. Y., Thiele, L., & **Ho, M.** · Submitted to *OJAp* · 2026 · [ADS]

[31] **`calypso`: a Parameter-Conditioned Stochastic Surrogate Model for Circumbinary Accretion Time-Series**
Siwek, M., **Ho, M.**, & Bellinger, E. · Submitted to *MNRAS* · 2026 · [ADS]

[30] **Learning the Universe: 3 h^{-1} Gpc Tests of a Field Level *N*-body Simulation Emulator**
Scoggins, M., **Ho, M.**, Villaescusa-Navarro, F., et al. · Submitted to *ApJ* · 2025 · [ADS]

[29] **Constructing impactful machine learning research for astronomy: Best practices for researchers and reviewers**
Huppenkothen, D., Ntampaka, M., **Ho, M.**, et al. · Submitted to *BAAS* · 2023 · [ADS]

PUBLISHED (28)

[28] **Agentic Systems for Calibrating Cosmological Simulations**
Tirapongprasert, C.† & **Ho, M.** · *ML and the Physical Sciences Workshop (ML4PS 2026)* · 2026 · [ADS]

[27] **Introducing sapphire: Towards Hybrid Physics-Informed, Data-Driven Modeling of Galaxy Formation**
Pandya, V., Bryan, G. L., …, **Ho, M.**, et al. · *ApJ*, 1011, 39 · 2026 · [ADS]

[26] **DegenDetector: Symbolic Recovery of Parameter Degeneracies in Bayesian Posteriors**
Tirapongprasert, C.† & **Ho, M.** · *Conference on Physics and AI (PAI 2026)* · 2026 · [ADS]

[25] **Learning at the Edge: Tailed-Uniform Sampling for Robust Simulation-Based Inference**
Tirapongprasert, C.† & **Ho, M.** · *Conference on Physics and AI (PAI 2026)* · 2026 · [ADS]

[24] **Learning the Universe: The Structure of Dust Attenuation Curves in Galaxy Simulations**
Sommovigo, L., Bartlett, D. J., …, **Ho, M.**, et al. · *ApJ*, 1008, 218 · 2026 · [ADS]

[23] **Flexible Simulation Based Inference for Galaxy Photometric Fitting with Synthesizer**
Harvey, T., Lovell, C., Newman, S., …, **Ho, M.**, et al. · *MNRAS*, 547, 1 · 2026 · [ADS]

[22] **Simulation-Based Inference for Direction Reconstruction of Ultra-High-Energy Cosmic Rays with Radio Arrays**
Macias, O., Mason, Z., **Ho, M.**, et al. · *Phys. Rev. D*, 113, 6 · 2026 · [ADS]

[21] **Reconstructing Galaxy Cluster Mass Maps using Score-based Generative Modeling**
Hsu, A.†, **Ho, M.**, Lin, J.†, et al. · *OJAp*, 8, 92 · 2025 · [ADS]

[20] **Learning the Universe: Cosmological and Astrophysical Parameter Inference with Galaxy Luminosity Functions and Colours**
Lovell, C., Starkenburg, T., **Ho, M.**, et al. · *MNRAS*, 544, 4 · 2025 · [ADS]

[19] **Creating halos with auto-regressive multi-stage networks**
Pandey, S., Modi, C., …, **Ho, M.**, et al. · *Phys. Rev. D*, 112, 10 · 2025 · [ADS]

[18] **Learning the Universe: physically-motivated priors for dust attenuation curves**
Sommovigo, L., Cochrane, R. K., …, **Ho, M.**, et al. · *ApJ*, 990, 114 · 2025 · [ADS]

★ [17] **Ordered Embeddings and Intrinsic Dimensionalities with Information-Ordered Bottlenecks**
Ho, M., Zhao, X., & Wandelt, B. · *Mach. Learn.: Sci. Technol.*, Volume 6, Issue 3 · 2025 · [ADS]

[16] **Cosmology with One Galaxy: Auto-Encoding the Galaxy Properties Manifold**
Lue, A., Genel, S., …, **Ho, M.**, et al. · *ApJ*, Volume 986, Issue 2 · 2025 · [ADS]

[15] **RTFAST-Spectra: Emulation of X-ray reverberation mapping for active galactic nuclei**
Ricketts, B. J., Huppenkothen, D., …, **Ho, M.**, et al. · *MNRAS*, 538, 2 · 2025 · [ADS]

[14] **Sensitivity Analysis of Simulation-Based Inference for Galaxy Clustering**
Modi, C., Pandey, M., **Ho, M.**, et al. · *MNRAS*, 536, 1 · 2025 · [ADS]

[13] **Bye-bye, Local-in-matter-density Bias: The Statistics of the Halo Field Are Poorly Determined by the Local Mass Density**
Bartlett, D., **Ho, M.**, & Wandelt, B. · *ApJ*, 977, 2 · 2024 · [ADS]

[12] **Inpainting Galaxy Counts onto *N*-Body Simulations over Multiple Cosmologies and Astrophysics**
Bourdin, A.†, Legin, R., **Ho, M.**, et al. · *ICML 2024 – AI4Science Workshop* · 2024 · [ADS]

[11] **Building Trustworthy Machine Learning Models for Astronomy**
Ntampaka, M., **Ho, M.**, & Nord, B. · *ADASS XXXI*, ASP Conf. Ser., 535, 123 · 2024 · [ADS]

[10] **Metallicity Dependence of Pressure-regulated Feedback-modulated Star Formation in the TIGRESS-NCR Simulation Suite**
Kim, C., Ostriker, E., …, **Ho, M.**, et al. · *ApJ*, 972, 1 · 2024 · [ADS]

★ [9] **LtU-ILI: An All-in-One Framework for Implicit Inference in Astrophysics and Cosmology**
Ho, M., Bartlett, D., Chartier, N., et al. · *OJAp*, 7 · 2024 · [ADS]

[8] **Posterior sampling of the initial conditions of the universe from non-linear large scale structures using score-based generative models**
Legin, R., **Ho, M.**, Lemos, P., et al. · *MNRAS*, 527, 1 · 2024 · [ADS]

[7] **Benchmarks and Explanations for Deep Learning Estimates of X-ray Galaxy Cluster Masses**
Ho, M., Soltis, J., Farahi, A., et al. · *MNRAS*, 524, 3 · 2023 · [ADS]

[6] **A Machine Learning Approach to Enhancing eROSITA Observations**
Soltis, J., …, **Ho, M.**, et al. · *ApJ*, 940, 60S · 2022 · [ADS]

★ [5] **The Dynamical Mass of the Coma Cluster from Deep Learning**
Ho, M., Ntampaka, M., Rau, M. M., …, Lansberry, A.[†], Ruehle, F.[†], et al. · *Nat. Astron.*, 6 (8), 936–941 · 2022 · [ADS]

[4] **CLMM: a LSST-DESC Cluster weak Lensing Mass Modeling library for cosmology**
Aguena, M., Avestruz, C., …, **Ho, M.**, et al. · *MNRAS*, 508, 6092 · 2021 · [ADS]

[3] **Approximate Bayesian Uncertainties on Deep Learning Dynamical Mass Estimates of Galaxy Clusters**
Ho, M., Farahi, A., Rau, M. M., & Trac, H. · *ApJ*, 908, 204H · 2021 · [ADS]

[2] **Aging haloes: implications of the magnitude gap on conditional statistics of stellar and gas properties of massive haloes**
Farahi, A., **Ho, M.**, & Trac, H. · *MNRAS*, 493, 1, 1361–1374 · 2020 · [ADS]

[1] **A Robust and Efficient Deep Learning Method for Dynamical Mass Measurements of Galaxy Clusters**
Ho, M., Rau, M. M., Ntampaka, M., et al. · *ApJ*, 887, 1 · 2019 · [ADS]

Mentoring

19 students (3 Ph.D., 5 Master's, 11 undergraduate) · 5 now in Ph.D. programs · 8 student-led papers

	STUDENT	PROGRAM	PAPERS	NOW
2026 –	**James Baldwin**	Ph.D., U. de Montréal		
2026 –	**Fatemeh Hafezianzadeh**	Ph.D., CMU		
2026 –	**Vedanti Kshirsagar**	Master's, CMU		
2026 –	**Anshul Kumar**	Master's, CMU	[36]	
2026 –	**Julia Liu**	Master's, CMU		
2026 –	**Yixi Zhao**	Master's, CMU		
2025 –	**Eddie Aljamal**	Ph.D., U. Michigan		
2025 –	**Ariyana Bonab**	Master's, CUNY		
2024 –	**Chaipat Tirapongprasert**	Undergrad, Columbia	[25, 26, 28, 33]	
2024 – 2025	**Thais Velazquez**	Undergrad, Barnard		
2023 – 2025	**Antoine Bourdin**	Undergrad, U. de Montréal	[12]	Ph.D. student, U. Chicago
2023 – 2025	**Kalvyn Adams**	Undergrad, CU Boulder	[34, 35]	Ph.D. student, UCLA
2022 – 2024	**Alan Hsu**	Undergrad, CMU	[21]	Ph.D. student, Harvard
2023 – 2024	**Caleb Ogle**	Undergrad, CU Boulder	[34, 35]	Ph.D. student, UW–Milwaukee
2022 – 2023	**Joyce Lin**	Undergrad, CMU	[21]	Ph.D. student, UW–Madison
2022 – 2023	**Kevin Hu**	Undergrad, Yale		
2021	**Bryant Dean**	Undergrad, Morehouse		
2020 – 2021	**Faith Ruehle**	Undergrad, CMU	[5]	
2019 – 2020	**Alexa Lansberry**	Undergrad, CMU	[5]	

Software Packages

2025 **ltu-cmass** maho3/ltu-cmass
A modular simulator of SDSS CMASS galaxy clustering.

2024 **ltu-ili** maho3/ltu-ili
An all-in-one framework package for machine learning inference in astrophysics and cosmology.

2024 **cuboid_remap_jax** maho3/cuboid_remap_jax
A Jax implementation to remap cubic volumes with periodic boundary conditions to non-cuboid volumes.

2024 **pytorch-iobs** maho3/pytorch-iobs
A lightweight implementation of Information-Ordered Bottlenecks (IOBs) in PyTorch.

2021 **CLMM** LSSTDESC/CLMM
A Python library for performing galaxy cluster mass reconstruction from weak lensing observables.

Presentations

INVITED (24)

Nov 2026	**NYU**, CCPP Astro Seminar	New York City, NY
Oct 2026	**IAS**, Bahcall Lunch	Princeton, NJ
Aug 2026	**CMU**, Machine Learning Department Seminar	Pittsburgh, PA
Aug 2026	**CMU**, McWilliams Seminar	Pittsburgh, PA
Apr 2026	**AMNH**, Astronomy Seminar	New York City, NY
Apr 2026	**Columbia University**, Astronomy Colloquium	New York City, NY
Feb 2026	**UToledo**, Physics Hot Topics Seminar	Toledo, OH
Feb 2026	**UToledo**, Physics Colloquium	Toledo, OH
Feb 2026	**UArizona**, Astronomy Colloquium	Tuscon, AZ
Feb 2026	**UT Austin**, Weinberg Institute Seminar	Austin, TX
Feb 2026	**UT Austin**, Physics Colloquium	Austin, TX
Jan 2026	**American University**, Physics Colloquium	Washington, DC
Jan 2026	**Illinois State University**, Physics Colloquium	Normal, IL
Jan 2026	**CCA**, Mock NYC: Improving Galaxy Modeling for Cosmological Inference	New York City, NY
Aug 2025	**Harvard University**, NSF IAIFI Summer Workshop	Cambridge, MA
Apr 2025	**Yale University**, Data Science and Physics Seminar	New Haven, CT
Mar 2025	**STScI**, Chesapeake ML-Astro Group	Baltimore, MD
Mar 2024	**University of Bristol**, SBI for Galaxy Evolution	Bristol, UK
Nov 2022	**IAP**, Univers Seminar	Paris, FR
Mar 2022	**Yale University**, Data Science X Astronomy & Astrophysics Seminar	New Haven, CT
Mar 2022	**Fermilab**, Cosmic Physics Center	Batavia, IL
Feb 2021	**CMU**, NSF AI Planning Institute for Physics of the Future	Pittsburgh, PA
Dec 2020	**DESI**, Artificial Intelligence Interest Group	Virtual
Jun 2019	**LMU Munich Observatory**, Weak Lensing Seminar	Munich, Germany

CONTRIBUTED (15)

Nov 2026	**CMU**, KAAI AI Methods for Turbulence and Early-Universe Signals	Pittsburgh, PA
Aug 2026	**CMU**, KAAI Foundation Models and Agentic Methods for Astrophysical Simulation	Pittsburgh, PA
Jul 2026	**UvA**, Cosmic Horizons Conference	Amsterdam, NL
Sep 2025	**NSF-Simons AI Institute for the Sky**, Open SkAI	Chicago, IL
May 2025	**UT Austin**, Cosmic Horizons Conference	Austin, TX
Dec 2024	**CCA**, Cosmology and Galaxy Astrophysics with Simulations and ML	New York City, NY
May 2024	**European AI For Fundamental Physics**, Annual Conference	Amsterdam, NL
Nov 2023	**IAP**, Debating the Potential of ML in Astronomical Surveys	Paris, FR
Apr 2022	**STScI**, Galaxy Clusters 2022: Challenging Our Cosmological Perspectives	Baltimore, MD
Sep 2021	**ESA**, A Multi-Wavelength View of Galaxy Clusters	Virtual
Jun 2020	**Joint Statistical Meeting**, Astrostatistics Student Paper Finalists	Philadelphia, PA
Jun 2019	**Artificial Intelligence Methods in Cosmology Workshop**	Ascona, Switzerland
May 2018	**CMU**, ML in Science and Engineering Conference	Pittsburgh, PA
Apr 2015	**NCSA**, Students Pushing Innovation Seminar	Urbana, IL
Dec 2014	**NCSA**, Students Pushing Innovation Seminar	Urbana, IL

Teaching Experience

2026	**Graduate Research Seminar II**, Guest Lecturer	Columbia
2025	**Astro-Skills: Numerical & Statistical**, Guest Lecturer	Columbia
2025	**IMPRS Heidelberg Summer School on AI for Astronomy**, Lecturer	Heidelberg
2024	**Frontiers of Astrophysics**, Guest Lecturer	Columbia
2021	**Introduction to Astronomy**, Instructor	CMU
2017 – 2021	**Teaching Assistant**, 9 course sections Matter & Interactions I (×3) · Physics II for Engineering and Physics Students (×3) · Physics I for Engineering Students · Physics for Future Presidents · Advanced Computational Physics	CMU

Organized Conferences & Workshops

2026	**KAAI Foundation Models & Agentic Methods for Astrophysical Simulation** Lead Organizer	Pittsburgh, PA
2025	**SBI for Galaxy Evolution Workshop**, SOC	Bristol, UK
2024	**Cosmology and Galaxy Astrophysics with Simulations and ML**, SOC	New York City, NY
2024	**SBI for Galaxy Evolution Workshop**, SOC	Bristol, UK

Service & Outreach

RESEARCH COMMUNITY & INSTITUTIONAL SERVICE

2025 – 2026	**Columbia Astro-ML Interest Group**, Organizer
2024 –	**Aquila Consortium Contact Unit**, Committee Member
2023	**Learning Learning the Universe**, Organizer and Presenter

2020 – 2021 **McWilliams Software Seminar Series**, Organizer

2020 – 2021 **Vera Computing System Commissioning Committee**, Committee Member

2017 – 2021 **Department of Physics Industry Speaker Seminar Series**, Founder, Organizer

2019 – 2020 **CMU Data Science Club**, Project Lead

PUBLIC OUTREACH

2025 **Astronomy on Tap**, Presenter — Baltimore

2025 **Columbia Public Outreach Science Talks**, Presenter — New York City

2022 **Astronomy on Tap**, Presenter — Pittsburgh

2021 **Carnegie Science Center**, Career Panelist — Pittsburgh

Peer Review & Memberships

REFEREE FOR

The Astrophysical Journal
Nature Astronomy
Journal of Cosmology and Astrophysics
Monthly Notices of the Royal Astronomical Society

MEMBER OF

Simons Collaboration on Learning the Universe
Euclid Consortium
Aquila Consortium
LSST-Dark Energy Science Collaboration
American Statistical Association